PAPER ID-310614

Roll No: | | | | | | | | | | | | | |

**B.TECH.**
**(SEM VII) THEORY EXAMINATION 2020-21**
**ARTIFICIAL INTELLIGENCE**

***Time: 3 Hours*** ***Total Marks: 70***

**Note: 1.** Attempt all Sections. If require any missing data; then choose suitably.

**SECTION A**

**1. Attempt *all* questions in brief. 2 x 7 = 14**

| Qno. | Question | Marks | CO |
|---|---|---|---|
| a) | What is heuristic function? | 2 | CO1 |
| b) | Write the difference between supervised and unsupervised learning. | 2 | CO4 |
| c) | List down the characteristics of agent. | 2 | CO1 |
| d) | List some of the uniform search technique. | 2 | CO2 |
| e) | Differentiate between forward and backward chaining. | 2 | CO3 |
| f) | What is bay's rule? | 2 | CO5 |
| g) | Define reinforcement learning. | 2 | CO4 |

**SECTION B**

**2. Attempt any *three* of the following: 7 x 3 = 21**

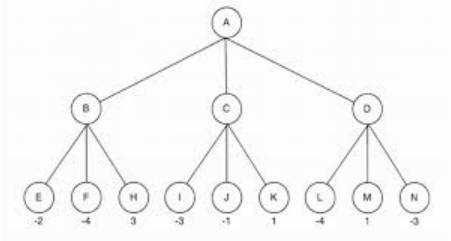
| Qno. | Question | Marks | CO |
|---|---|---|---|
| a. | Explain DFS algorithm with suitable example. | 7 | CO2 |
| b. | Define a well-formed formula (wff) and List some of the rules of inference. | 7 | CO3 |
| c. | What are Statistical learning models? Show with suitable example. | 7 | CO4 |
| d. | Define PCA. Differentiate between Principle Component Analysis (PCA) and Linear Discriminant Analysis (LDA) | 7 | CO5 |
| e. | Explain state space approach for solving any AI problem. | 7 | CO1 |

**SECTION C**

**3. Attempt any *one* part of the following: 7 x 1 = 7**

| Qno. | Question | Marks | CO |
|---|---|---|---|
| a. | Describe the four categories under which AI is classified with examples. | 7 | CO1 |
| b. | List various components of natural language understanding process. Describe syntactic analysis and semantic analysis in brief. | 7 | CO1 |

**4. Attempt any *one* part of the following: 7 x 1 = 7**

| Qno. | Question | Marks | CO |
|---|---|---|---|
| a. | Explain Alpha-Beta pruning? Solve the following question-  | 7 | CO2 |

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| | | | |
|---|---|---|---|
| b. | Discuss Constraint Satisfaction problem with an algorithm for solving a Cryptarithmetic problem<br>BASE<br>+ BALL<br>GAMES | 7 | CO2 |

**5. Attempt any *one* part of the following: 7 x 1 = 7**

| Qno. | Question | Marks | CO |
|---|---|---|---|
| a. | Explain resolution in predicate logic with suitable example. | 7 | CO3 |
| b. | Trace the operation of the unification algorithm on each of the following pairs of literals:<br>I. f(Marcus) and f(Caesar)<br>II. f(x) and f(g(y))<br>III. f(Marcus, g(x, y)) and f(x, g(Caesar, Marcus)). | 7 | CO3 |

**6. Attempt any *one* part of the following: 7 x 1 = 7**

| Qno. | Question | Marks | CO |
|---|---|---|---|
| a. | Define decision tree? Explain it's with suitable example. | 7 | CO4 |
| b. | How can use Expectation-Maximization (EM Algorithm) in machine learning? Explain with appropriate example. | 7 | CO4 |

**7. Attempt any *one* part of the following: 7 x 1 = 7**

| Qno. | Question | Marks | CO |
|---|---|---|---|
| a. | Give the block diagram of pattern recognition system. Explain in brief. | 7 | CO5 |
| b. | What do you mean by support vector machine (SVM)? Explain in detail with suitable example. | 7 | CO5 |

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